Number

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In a class of 30 students, 15 students play football and 20 students play basketball. Every student plays at least one of the two sports. Using a Venn diagram, find the number of students who play both football and basketball.

Let F be the set of students who play football and B be the set of students who play basketball. We are given that the total number of students is 30. The number of students who play football is $|F| = 15$. The number of students who play basketball is $|B| = 20$. Since every student plays at least one of the two sports, the union of the two sets is equal to the total number of students: $|F \cup B| = 30$. We can use the principle of inclusion-exclusion for two sets, which states.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A number line is shown below. What is the value of the number indicated by the arrow?

The number line shows integers from -5 to 7. The arrow points directly to the mark labelled '0'. Therefore, the value of the number indicated by the arrow is 0. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The histogram shows the distribution of scores for a group of students in a mathematics test. The total number of students who scored between 0 and 40 is 32. The frequency density for the class interval $20 \leq x < 40$ is 0.8. Calculate the number of students who scored between 20 and 40.

The area of a bar in a histogram represents the frequency of the corresponding class interval. The area is calculated by frequency density multiplied by the class width. The class interval $20 \leq x < 40$ has a width of $40 - 20 = 20$. The frequency density for this interval is given as 0.8. Therefore, the frequency (number of students) in this interval is calculated as: Frequency = Frequency Density $\times$ Class Width Frequency = 0.8×20 Frequency = 16 Thus, 16 students scored between 20 and 40.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows a number line. The distance between the points marked A and B is 15 units. The coordinate of A is -7. Find the possible coordinates of B.

Let the coordinate of point B be x. The distance between two points on a number line is the absolute difference of their coordinates. Therefore, the distance between A and B is $|x-(-7)| = |x+7|$. We are given that this distance is 15 units. So, we have the equation $|x+7| = 15$. This gives us two possibilities: $x+7 = 15$ or $x+7 = -15$. Solving the first equation, $x = 15-7 = 8$. Solving the second equation, $x = -15-7 = -22$. Therefore, the possible coordinates of B are -22 and 8.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag contains n marbles, of which 4 are green. Two marbles are taken at random without replacement. If the probability that both are green is $\frac{2}{15}$, find the total number of marbles n.

$P(\text{both green}) = \frac{4}{n} \times \frac{3}{n-1} = \frac{12}{n(n-1)}$. Set equal to $\frac{2}{15}$: $\frac{12}{n(n-1)} = \frac{2}{15} \Rightarrow 2n(n-1) = 180 \Rightarrow n(n-1) = 90 \Rightarrow n^2 - n - 90 = 0 \Rightarrow (n-10)(n+9) = 0$. Since $n > 0$, $n = 10$. [M1 for setting up probability equation, M1 for quadratic n(n-1)=90, A1 for 10]

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Set A has 12 elements and set B has 8 elements. If $A \cap B eq \varnothing$, what is the maximum possible number of elements in $A \cup B$?

Formula: $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 12 + 8 - n(A \cap B) = 20 - n(A \cap B)$. To maximize the union, we must minimize the intersection. Since $A \cap B eq \varnothing$, the minimum possible value of $n(A \cap B)$ is 1. Thus maximum $n(A \cup B) = 20 - 1 = 19$. [M1 for maximizing union by minimizing intersection to 1, A1 for 19]

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $y = x^2 - 4x + 3$ is shown. The graph of the linear function $y = x - 1$ is also shown. The graphs intersect at points A and B. Find the coordinates of point A.

To find the intersection points, we set the two equations equal to each other: $x^2 - 4x + 3 = x - 1$. Rearranging the terms to form a quadratic equation: $x^2 - 5x + 4 = 0$. Factoring the quadratic equation: $(x-1)(x-4) = 0$. This gives us two possible x-values for the intersection points: $x = 1$ or $x = 4$. For $x = 1$, we can find the corresponding y-coordinate using the linear equation $y = x - 1$: $y = 1 - 1 = 0$. So, one intersection point is $(1, 0)$. For $x = 4$, we can find the corresponding y-coordinate.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows the graph of the quadratic function $f(x) = x^2 + ax + b$. The graph passes through the points $(-1, 8)$ and $(3, 0)$. Find the values of a and b.

The function is given by $f(x) = x^2 + ax + b$. Since the graph passes through $(-1, 8)$, we have $f(-1) = 8$. Substituting $x = -1$ into the function: $(-1)^2 + a(-1) + b = 8$ $1 - a + b = 8$ $-a + b = 7$ (*) Since the graph passes through $(3, 0)$, we have $f(3) = 0$. Substituting $x = 3$ into the function: $(3)^2 + a(3) + b = 0$ $9 + 3a + b = 0$ $3a + b = -9$ (**) Now we have a system of two linear equations with two variables: 1) $-a + b = 7$ 2) $3a + b = -9$ Subtract equation (1) from equation (2).

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the right-angled triangle ABC, $\angle ABC = 90^\circ$. If $AB = 5$ cm and $BC = 12$ cm, find the length of the hypotenuse AC.

The triangle ABC is a right-angled triangle with $\angle ABC = 90^\circ$. We are given the lengths of the two shorter sides, AB = 5 cm and BC = 12 cm. We can use the Pythagorean theorem to find the length of the hypotenuse AC. The Pythagorean theorem states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. $AC^2 = AB^2 + BC^2$. Substituting the given values: $AC^2 = 5^2 + 12^2$. $AC^2 = 25 + 144$. $AC^2 = 169$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a right-angled triangle ABC, with the right angle at B. The length of side AB is 10 cm and the length of side BC is 24 cm. A semicircle is drawn with AC as its diameter. Calculate the area of the semicircle in terms of π.

First, we need to find the length of the hypotenuse AC using the Pythagorean theorem: $AC^2 = AB^2 + BC^2$. $AC^2 = 10^2 + 24^2 = 100 + 576 = 676$. So, $AC = \sqrt{676} = 26$ cm. The radius of the semicircle is half the diameter AC, so the radius $r = \frac{26}{2} = 13$ cm. The area of a semicircle is given by $\frac{1}{2}\pi r^2$. Substituting the radius, we get $\frac{1}{2}\pi(13)^2 = \frac{1}{2}\pi(169) = 84.5\pi$ cm^2. Common error: Using the sides AB or BC as diameter, or for.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows the graph of the quadratic function $y = x^2 + bx + c$. The graph intersects the x-axis at $(1, 0)$ and $(5, 0)$. Find the values of b and c.

Since the graph intersects the x-axis at $(1, 0)$ and $(5, 0)$, these are the roots of the quadratic equation $x^2 + bx + c = 0$. We can use Vieta's formulas, which state that for a quadratic equation $ax^2 + bx + c = 0$, the sum of the roots is $-b/a$ and the product of the roots is c/a. In this case, $a = 1$. Sum of roots: $1 + 5 = 6$. Therefore, $-b/1 = 6$, which means $b = -6$. Product of roots: $1 \times 5 = 5$. Therefore, $c/1 = 5$, which means $c = 5$. Thus, the values are $b = -6$ and $c = 5$.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a right-angled triangle, one of the acute angles is 30° and the side opposite to this angle is 6 cm. Calculate the length of the hypotenuse.

Let the right-angled triangle be ABC, with the right angle at C. Let angle A be 30°. The side opposite to angle A is BC, which is given as 6 cm. The hypotenuse is AB. We use the sine function: $\sin(A) = \frac{\text{opposite}}{\text{hypotenuse}}$. So, $\sin(30^\circ) = \frac{BC}{AB}$. We know that $\sin(30^\circ) = \frac{1}{2}$. Therefore, $\frac{1}{2} = \frac{6}{AB}$. Solving for AB, we get $AB = 2 \times 6 = 12$ cm.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A rectangular prism has a length of 5 cm, a width of 3 cm, and a height of 4 cm. Calculate the volume of the prism.

The volume of a rectangular prism is calculated by the formula: Volume = length × width × height. Given: Length = 5 cm Width = 3 cm Height = 4 cm Volume = 5 cm × 3 cm × 4 cm Volume = 15 cm^2 × 4 cm Volume = 60 cm^3 Therefore, the volume of the prism is 60 cm^3.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $y = f(x)$ is shown below. Find the value of $f(0)$.

The graph represents the function $y = f(x)$. To find the value of $f(0)$, we need to find the y-coordinate of the point where the graph intersects the y-axis. The y-axis is where $x = 0$. Looking at the graph, when $x = 0$, the corresponding y-value is 3. Therefore, $f(0) = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

The diagram shows the graph of the quadratic function $y = x^2 - 4x + k$. The graph intersects the x-axis at point P(1, 0) and the y-axis at point Q. Determine the value of k.

The graph of the quadratic function $y = x^2 - 4x + k$ passes through the point P(1, 0). Substituting the coordinates of P into the equation, we get: $0 = (1)^2 - 4(1) + k$ $0 = 1 - 4 + k$ $0 = -3 + k$ $k = 3$. Therefore, the value of k is 3.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Transformations & Vectors

In the diagram below, lines AB and DE are parallel. Points A, C, and D are collinear, and points B, C, and E are collinear. If $AC = 6$ cm, $CD = 9$ cm, and $DE = 12$ cm, find the length of AB.

Since AB is parallel to DE, we can identify two similar triangles: ΔABC and ΔDEC. This is due to the property of alternate interior angles being equal when lines are parallel, and vertically opposite angles at C being equal. Specifically, $\angle BAC = \angle CDE$ and $\angle ABC = \angle DEC$ (alternate interior angles). Also, $\angle ACB = \angle DCE$ (vertically opposite angles). Because the triangles are similar, the ratio of their corresponding sides is equal.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Probability & Statistics

A point $A(3, 5)$ is mapped to $A'(1, 3)$ by a single transformation. The same transformation is then applied to a second point $B(x, y)$, mapping it to $B'(x - 2, y - 2)$. The vector representing the displacement from A to A' is $\vec{v}$. A point $C(2, 1)$ is reflected in the line $y = x$ to give point C'. Calculate the magnitude of the vector $A\vec{C}'$.

The transformation that maps $A(3, 5)$ to $A'(1, 3)$ is a translation. The translation vector $\vec{v}$ can be found by subtracting the coordinates of A from A': $\vec{v} = A' - A = (1 - 3, 3 - 5) = (-2, -2)$. This means the translation moves each point 2 units to the left and 2 units down. The problem states that point $B(x, y)$ is mapped to $B'(x - 2, y - 2)$ by the same transformation, which confirms our translation vector is correct. Next, point $C(2, 1)$ is reflected in the line $y = x$. A reflection in the line.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Functions

A survey of 100 students about their preferred social media platforms yielded the following information: - 55 students use Platform A. - 62 students use Platform B. - 18 students use neither Platform A nor Platform B. - x students use both Platform A and Platform B. Find the value of x.

Let A be the set of students who use Platform A and B be the set of students who use Platform B. We are given the total number of students is 100. Number of students who use neither A nor B is 18. Therefore, the number of students who use at least one platform is $100 - 18 = 82$. This means $|A \cup B| = 82$. We are given $|A| = 55$ and $|B| = 62$. The formula for the union of two sets is $|A \cup B| = |A| + |B| - |A \cap B|$. Here, $|A \cap B|$ represents the number of students who use both platforms, which is giv.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A triangle ABC has vertices A(2, 3), B(5, 1), and C(3, 5). This triangle is subjected to a sequence of transformations. First, it is reflected in the line $y = x$ to produce triangle A'B'C'. Then, triangle A'B'C' is translated by the vector $(-3; 1)$ to produce triangle A''B''C''. ; Determine the coordinates of the vertices of triangle A''B''C''.

Let the original triangle ABC have vertices A(-2, 2), B(1, 3), and C(-1, 1). Step 1: Reflection in the line $y = x$. When a point (x, y) is reflected in the line $y = x$, its image is (y, x). Applying this to triangle ABC: A(-2, 2) becomes A'(2, -2). B(1, 3) becomes B'(3, 1). C(-1, 1) becomes C'(1, -1). Step 2: Translation of triangle A'B'C'. Triangle A'B'C' is translated by the vector $(-3; 1)$. This means we subtract 3 from the x-coordinate and add 1 to the y-coordinate of each vertex. Applying th.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Algebra & Graphs

The diagram shows a point A and its image A' after a single transformation. The coordinates of A are (2, 1) and the coordinates of A' are (5, 3). The transformation can be represented by the vector $(p; q)$. Find the value of p.

The transformation from point A to its image A' can be represented by a translation vector. The change in the x-coordinate is the difference between the x-coordinate of A' and the x-coordinate of A. The change in the y-coordinate is the difference between the y-coordinate of A' and the y-coordinate of A. The translation vector is given by $(x_{A'} - x_A; y_{A'} - y_A)$. In this case, $x_A = 2$, $y_A = 1$, $x_{A'} = 5$, and $y_{A'} = 3$. Therefore, the translation vector is.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in estimation, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in rounding, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in four operations with whole numbers, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for estimation consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for rounding consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for order of operations with whole numbers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in order of operations with whole numbers, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for using technology for operations with whole numbers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
For the model $f(x) = 6x + 7$ in using technology for operations with whole numbers, evaluate $f(4)$.

a. 31 ✓
b. 37
c. 24
d. 35

Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
For the model $f(x) = 2x + 8$ in check reasonableness of calculations, evaluate $f(3)$.

a. 14 ✓
b. 16
c. 6
d. 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Find the lowest common multiple (LCM) of 12 and 15.

a. 30
b. 45
c. **60** ✓
d. 90

Prime factorisation: $12 = 2^2 \times 3$, $15 = 3 \times 5$. LCM $= 2^2 \times 3 \times 5 = 4 \times 15 = 60$. [B1]

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)
Universal set $\mathscr{E} = \{x : x \text{ is an integer}, 1 \leq x \leq 20\}$. $P = \{x : x \text{ is a prime number}\}$ and $O = \{x : x \text{ is an odd number}\}$. What is $n(P \cap O')$?

a. **1** ✓
b. 2
c. 8
d. 9

The prime numbers in $\mathscr{E}$ are $\{2, 3, 5, 7, 11, 13, 17, 19\}$. The set O' consists of all even numbers. The only even prime number is 2, so $P \cap O' = \{2\}$, which has exactly 1 element: $n(P \cap O') = 1$. [M1 for identifying even prime 2, A1 for 1]

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows a number line with points marked. What is the value of x?

a. -5

b. **-3** ✓

c. 1

d. 5

The number line shows integer values from -6 to 6. Point x is located exactly at the mark for -3. Therefore, the value of x is -3.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Universal set $\mathscr{E} = \{x : x \in \mathbb{Z}^+, 1 \le x \le 20\}$. Set $P = \{x : x \text{ is a prime number}\}$. What is $n(P)$?

a. **8** ✓

b. 7

c. 9

d. 10

Primes up to 20 are $\{2, 3, 5, 7, 11, 13, 17, 19\}$, so $n(P) = 8$. [B1]

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Universal set $\mathscr{E} = \{x : x \in \mathbb{Z}^+, x \le 15\}$. Set $A = \{x : x \text{ is a multiple of } 3\}$ and $B = \{x : x \text{ is a multiple of } 5\}$. Find $A \cap B$.

a. $\{15\}$ ✓

b. $\{3, 5, 15\}$

c. $\{3, 5, 6, 9, 10, 12, 15\}$

d. $\varnothing$

Multiples of both 3 and 5 are multiples of 15: only $\{15\}$. [B1]

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a class of 60 students, all students study at least one of the subjects French (F) or Spanish (S). The number of students who study French is $n(F)$, the number of students who study Spanish is $n(S)$, and the number of students who study both French and Spanish is $n(F \cap S)$. Given that $n(F) = 3x - 4$, $n(S) = 2x + 6$, and $n(F \cup S) = 50$, find the value of x.

a. 4

b. 5

c. **6 ✓**

d. 7

Let $n(F)$ be the number of students studying French, and $n(S)$ be the number of students studying Spanish. Let $n(F \setminus S)$ be the number of students studying only French, and $n(S \setminus F)$ be the number of students studying only Spanish. Let $n(F \cap S)$ be the number of students studying both subjects. We are given: $n(F) = 3x - 4$ $n(S) = 2x + 6$ $n(F \setminus S) = 14$ $n(S \setminus F) = 18$ We know that the total number of students studying a subject is the sum of those studying only that subjec.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Functions

In a survey of 100 students about their preferred social media platforms, it was found that 60 use Platform A, 55 use Platform B, and 30 use both Platform A and Platform B. A further 15 students use neither Platform A nor Platform B. Let A be the set of students who use Platform A, and B be the set of students who use Platform B. (a) Represent this information on a Venn diagram. (b) Calculate the number of students who use only Platform A. (c) Calculate the number of students who use only Platform B. (d) If a student is chosen at random from the survey, what is the probability that they use Platform A or Platform B (or both)?

a. 75/100

b. **85/100 ✓**

c. 95/100

d. 70/100

Let $|U|$ be the total number of students surveyed, so $|U| = 100$. Let $|A|$ be the number of students who use Platform A, so $|A| = 60$. Let $|B|$ be the number of students who use Platform B, so $|B| = 55$. Let $|A \cap B|$ be the number of students who use both Platform A and Platform B, so $|A \cap B| = 30$. The number of students who use neither Platform A nor Platform B is 15. (a) To represent this on a Venn diagram: - The intersection $|A \cap B| = 30$. - The number of students who use only Platform A is.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A special sequence of numbers is defined by the recurrence relation $a_{n+1} = \frac{2a_n + k}{a_n}$, where k is a positive constant. The first term is $a_1 = 3$. If $a_3 = \frac{13}{5}$, find the value of k and the value of a_4.

a. $k = 3, a_4 = 7/3$

b. $k = 4, a_4 = 13/4$ ✓

c. $k = 5, a_4 = 9/5$

d. $k = 2, a_4 = 10/3$

We are given the recurrence relation $a_{n+1} = \frac{2a_n + k}{a_n}$, with $a_1 = 3$ and $a_3 = \frac{16}{5}$. First, we find a_2 in terms of k: $a_2 = \frac{2a_1 + k}{a_1} = \frac{2(3) + k}{3} = \frac{6 + k}{3}$. Next, we find a_3 in terms of k: $a_3 = \frac{2a_2 + k}{a_2} = \frac{2\left(\frac{6+k}{3}\right) + k}{\frac{6+k}{3}}$. To simplify, multiply the numerator by 3: $a_3 = \frac{2(6 + k) + 3k}{6 + k} = \frac{12 + 2k + 3k}{6 + k} = \frac{12 + 5k}{6 + k}$. We are given that $a_3 = \frac{16}{5}$. So, we set up th.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Algebra & Graphs

Rationalise the denominator and simplify: $\frac{10}{3 - \sqrt{5}}$.

a. $\frac{15 + 5\sqrt{5}}{2}$ ✓

b. $5(3 + \sqrt{5})$

c. $\frac{10(3 - \sqrt{5})}{4}$

d. $3 + \sqrt{5}$

Multiply numerator and denominator by the conjugate $3 + \sqrt{5}$:

$\frac{10(3 + \sqrt{5})}{(3 - \sqrt{5})(3 + \sqrt{5})} = \frac{10(3 + \sqrt{5})}{9 - 5} = \frac{10(3 + \sqrt{5})}{4} = \frac{5(3 + \sqrt{5})}{2} = \frac{15 + 5\sqrt{5}}{2}$. [M1 for multiplying by conjugate, M1 for difference of squares denominator, A1 for simplified form]

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Coordinate Geometry

A car travels a distance of 150 km, correct to the nearest 10 km, in a time of 2 hours, correct to the nearest 0.1 hour. What is the upper bound of the average speed in km/h?

a. **79.5 km/h** ✓

b. 77.5 km/h

c. 75.0 km/h

d. 81.6 km/h

Upper bound of speed = $\frac{\text{Upper bound of distance}}{\text{Lower bound of time}}$. Upper distance = $150 + 5 = 155$ km. Lower time = $2.0 - 0.05 = 1.95$ hours. Maximum speed = $\frac{155}{1.95} \approx 79.487$ km/h ≈ 79.5 km/h. [M1 for upper dist / lower time, A1 for 79.5]

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Mensuration

A bank account pays compound interest of 3% per year. How many complete years will it take for an initial deposit of $x3000$?

a. 12 years

b. 13 years

c. **14 years ✓**

d. 15 years

We need $2000(1.03)^n > 3000 \Rightarrow (1.03)^n > 1.5$. Taking logarithms: $n\ln(1.03) > \ln(1.5) \Rightarrow n > \frac{\ln(1.5)}{\ln(1.03)} \approx \frac{0.405465}{0.029559} \approx 13.72$. The minimum number of complete years is 14. [M1 for setting up inequality, A1 for 14 years]

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Geometry

Convert $0.1\dot{4}\dot{8}$ to a fraction in its simplest form.

a. $\frac{148}{1000}$

b. $\frac{49}{330}$ ✓

c. $\frac{37}{250}$

d. $\frac{7}{45}$

Let $x = 0.1484848\ldots$. Then $10x = 1.484848\ldots$ and $1000x = 148.484848\ldots$. Subtract: $1000x - 10x = 148 - 1 = 147 \Rightarrow 990x = 147 \Rightarrow x = \frac{147}{990}$. Divide numerator and denominator by 3: $\frac{49}{330}$. [M1 for 1000x - 10x, A1 for 49/330]

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

A rectangle has length $l = 8.4$ cm correct to 1 decimal place and width $w = 5.6$ cm correct to 1 decimal place. Calculate the lower bound for the area of the rectangle.

a. 47.04 cm²

b. **46.3425 cm² ✓**

c. 46.75 cm²

d. 47.7425 cm²

Lower bound of length = $8.4 - 0.05 = 8.35$ cm. Lower bound of width = $5.6 - 0.05 = 5.55$ cm. Lower bound of area = $8.35 \times 5.55 = 46.3425$ cm^2. [B1 for bounds 8.35 and 5.55, M1 for multiplying, A1 for 46.3425]

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Transformations & Vectors

Find the highest common factor (HCF) of 36 and 54.

a. 6

b. 9

c. **18** ✓

d. 27

Prime factorisation: $36 = 2^2 \times 3^2$ and $54 = 2 \times 3^3$. The HCF takes the lowest power of common factors: $\text{HCF} = 2^1 \times 3^2 = 2 \times 9 = 18$. [M1 for prime factors, A1 for 18]

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